RR-S-615C
September 4, 1969
SUPERSEDING
Fed. Spec. RR-S-615B
August 31, 1965

FEDERAL SPECIFICATION

SPONGES, METAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metal sponges (see 6.1).

1.2 Classification. Metal sponges shall be of the following types and classes as specified (see 6.2):

Type I - Corrosion resisting steel.
  Class A - Composition 430.
  Class B - Composition 434A.
Type II - Nickel-copper alloy.
Type III - Copper-nickel-zinc alloy.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-C-586 - Copper-Nickel-Zinc Alloy; Rod, Shapes and Flat Products with Finished Edges (Flat Wire, Strip, and Bar).
QQ-N-281 - Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-W-423 - Wire, Steel, Corrosion-Resisting.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Set-Up.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container grade) and Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestics Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:10Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publications:

F219-67 - Testing Fine Round and Flat Wire for Electronic Devices and Lamps.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

National Classification Board:

National Motor Freight Classification.

Issued by National Motor Freight Traffic Association, Inc., Agent

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee Agency, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), before production is commenced, a sample metal sponge of the type and class covered by this specification, shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product. Metal sponges delivered under this specification shall be the manufacturers' standard commercial product, except, for any changes necessary to meet specification requirements.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for his metal sponge provided the completed item complies with all provisions of this specification.

3.3.1 Wire, steel, fine, corrosion-resisting. Corrosion-resisting fine steel wire shall conform to 3.3.1.1 or 3.3.1.2, as applicable and shall be cold worked. When round cold worked wire is used, it shall have a minimum tensile strength of 135,000 psi. There shall be no evidence of tensile strength failure when tested as specified in 4.4.

3.3.1.1 Composition 430. In addition to the requirements in 3.3.1, corrosion-resisting fine steel wire shall conform to form I, composition 430, of QQ-W-423.

3.3.1.2 Composition 434A. In addition to the requirements in 3.3.1, type 434A corrosion-resisting fine steel wire shall conform to commercial quality as normally used in the metal sponge industry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:10Z
Check the source to verify that this is the current version before use.

3.3.2 Wire, nickel-copper-alloy. Nickel-copper-alloy wire shall conform to class A, form 7 of QQ-N-281.

3.3.3 Wire, Copper-nickel-zinc alloy. Copper-nickel-zinc alloy wire shall conform to all alloys, cold drawn, hard temper of QQ-C-586.

3.4 Construction. The metal sponge shall consist of not more than two continuous smooth edge ribbons curled under tension into an unbroken series of coils simultaniously formed under released tension into an interlocking cylindrical shape. The thickness or diameter for the smooth edge ribbons shall be 0.0015 plus or minus 0.0001 inch, prior to fabrication. Type I, classes A and B metal sponges shall be fabricated from corrosion-resisting steel specified in 3.3.1.1 and 3.3.1.2, respectively. The type II metal sponge shall be fabricated of nickel-copper-alloy specified in 3.3.2.

3.5 Dimensions. Each metal sponge shall be 4-1/2 inches plus or minus 1/2 inch in diameter and not less than 1-3/4 inches in thickness.

3.6 Weight. Each metal sponge shall weigh not less than 2-1/4 ounces.

3.7 Finish. *Metal sponges shall be clean and free from oil, grease, dirt, and areas of corrosion.*

3.8 Workmanship. The finished metal sponges shall not contain any rough or sharp edges, burrs, slivers or loose particles such as small pieces of wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When a preproduction sample is required, it shall be examined for defects listed in table I, dimensions and weight specified, and tested as specified in 4.4. Failure to pass the test in 4.4 shall be cause for rejection of the preproduction sample, the presence of any defects listed in table I or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, *except where otherwise indicated hereinafter.*

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Corrosion-resisting steel tensile strength. The tensile strength of the corrosion-resisting fine steel wire specified in 3.3.1 shall be tested as specified in 4.4. The lot shall be the amount of wire offered at one time. Three samples shall be taken from each lot of corrosion-resisting fine steel wire. Each sample shall be a minimum of 12 inches long and each taken from a different coil or reel. Failure of any sample to comply shall be cause for rejection of the lot.

4.3.2 End item inspection. The inspection lot shall be the number of metal sponges of one type and class offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated metal sponge.

4.3.2.1 Visual examination. Examination shall be made of the end item for the defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:10Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not clean.- - - - - - - - - - - - - - - - -	X	
	Any evidence of oil, grease or dirt.- - -	X	
	Area of corrosion.- - - - - - - - - - - -		X
Construction and workmanship	More than two continuous ribbons- - - - -		X
	Ribbon(s) not smooth edge or not curled into continuous series of helical coils.-	X	
	Ribbon(s) not formed into an interlocking cylindrical shape. - - - - - - - - - - - -	X	
	Any rough or sharp edges, burrs, slivers or loose particles.- - - - - - - - - - - -	X	

4.3.2.2 Dimensional examination. Inspection shall be made of the sample unit to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

4.3.2.3 Weight examination. Examination shall be made of the metal sponges for defects in weight. Any metal sponge failing to meet the weight requirements as specified in 3.6 shall be classified as a defect. The inspection level shall be S-4 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall consist of the number of containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Marking, exterior and interior	Omitted; incorrect; illegible, of improper size, location, sequence, or method of application.
Material	Component missing or damaged.
Contents	Number per container more or less than required.

4.3.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing and strapping shall be in accordance with PPP-B-636.

4.4 Tensile test. Conformance for the tensile strength specified in 3.3.1 shall be tested in accordance with ASTM F219-67.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.. Twelve metal sponges shall be packaged in a paperboard tube or sleeve, or in a paperboard or fiberboard box. The paperboard or fiberboard box shall conform to PPP-B-566, PPP-B-676, or PPP-B-636. The paperboard tube or sleeve shall be fabricated from a bending grade paperboard having a thickness of not less than 0.012 inch. The tubes or sleeves shall be designed to retain the metal sponges without spilling until individually removed for use. Care shall be exercised that sponges are not compressed to such a degree as to cause them to entangle with each other.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:10Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. The metal sponges shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. One hundred and forty-four metal sponges, packaged as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with appendix of PPP-B-636.

5.2.2 Level B. One-hundred and forty-four metal sponges, packaged as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.3 Level C. Metal sponges, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules, as applicable..

5.3 Palletization (Military requirements). Unless otherwise specified (see 6.2) metal sponges packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-129.

6. NOTES

6.1 Intended use. Metal sponges are intended to be used in galleys, kitchens and elsewhere for cleaning and scouring various items such as pots, pans, cooking utensils, sinks, and tables. Type I metal sponges may be considered as longer lasting under heavy uses.

6.2 Ordering data. Purchasers should select the preferred options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class of metal sponge required (see 1.2).
(c) First article. Whether preproduction sample is waived (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is not required (see 5.3).

6.3 For civil agency procurement. When level B packaging is required for civil agency procurement, the level A requirements in 5.1.1 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:10Z
Check the source to verify that this is the current version before use.

MILITARY CUSTODIANS:

Army - GL
Air Force - 84

Review activities:

Army - GL, MD
Air Force - 85

User activities:

Navy - SA
Air Force - 26

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

AGR
COM
DC
GSA
INT
JUS
PO
TR

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:10Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Sponges, Metal RR-S-615C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:10Z
Check the source to verify that this is the current version before use.